

The REN Formula: Construction, Properties, and Application in Hutchinson Synthesis

Contents

1	Introduction	2
2	The Base REN Formula	2
2.1	Definition and Parameters	2
2.2	Properties	2
2.3	Geometric Interpretation	2
3	Extending REN with Angular Mobility	2
3.1	Limitations of the Base Form	2
3.2	Simple Warp (Without Integrals)	3
3.3	Proportional Reparameterisation (CDF)	3
3.4	General REN Formula	3
3.5	Interpretation of the Extended Parameters	3
3.6	Limit Behaviour	3
4	Applications	3
4.1	General Applications	3
4.2	REN in Hutchinson Synthesis	3
4.2.1	Role in the Synthesiser	4
4.2.2	From One Curve to One Flower per Niche	4
4.2.3	Magnet and Guaranteed Clamp	4
4.2.4	A Note on the Codebase	4
5	Conclusion	5

1 Introduction

This document presents a family of polar curves named **REN (Radial Exponential Normalised)**, designed with the following goals:

- maximum formal simplicity
- automatic normalisation
- high geometric expressiveness

The formula evolves from a fully symmetric base version to deformed, locally controllable shapes.

2 The Base REN Formula

The fundamental formula is:

$$r(\theta) = \exp(-\lambda |\sin(k\theta)|^q) \quad (1)$$

2.1 Definition and Parameters

- $k \in \mathbb{N}^+$: number of lobes
- $\lambda > 0$: depth of the minima
- $q > 0$: shape control (sharpness)

2.2 Properties

- $r_{max} = 1$ (normalisation)
- periodicity: $\frac{2\pi}{k}$
- full symmetry

2.3 Geometric Interpretation

The function $\sin(k\theta)$ defines an angular distance from the symmetry points. The exponential introduces a controlled decay.

3 Extending REN with Angular Mobility

3.1 Limitations of the Base Form

The base formula has:

- a uniform distribution of lobes
- no possibility of local clustering

To overcome these limitations, we introduce a function $\phi(\theta)$ and define:

$$r(\theta) = \exp(-\lambda |\sin(k\phi(\theta))|^q) \quad (2)$$

3.2 Simple Warp (Without Integrals)

$$\phi(\theta) = \theta + \mu \sin(\theta - \theta_0) \quad (3)$$

Parameters: θ_0 is the cluster position, μ the deformation intensity. This simple warp does not guarantee proportional redistribution and may introduce global distortion.

3.3 Proportional Reparameterisation (CDF)

To redistribute angular space without distortion, we define a density:

$$w(t) = 1 + \mu e^{\kappa \cos(t - \theta_0)} \quad (4)$$

$$F(\theta) = \int_0^\theta w(t) dt \quad (5)$$

$$\phi(\theta) = \frac{2\pi}{F(2\pi)} F(\theta) \quad (6)$$

3.4 General REN Formula

$$r(\theta) = \exp(-\lambda |\sin(k\phi(\theta))|^q) \quad (7)$$

3.5 Interpretation of the Extended Parameters

- θ_0 : cluster center
- μ : clustering intensity
- κ : cluster width

3.6 Limit Behaviour

- $\mu = 0 \Rightarrow$ classic REN
- $\kappa \rightarrow \infty \Rightarrow$ point cluster

4 Applications

4.1 General Applications

- generation of natural shapes
- parametric design
- biological modelling

4.2 REN in Hutchinson Synthesis

REN was introduced as the geometric constraint used to shape the spatial dimension of **Hutchinson Synthesis**, a parametric additive synthesis system organising sound events over an N -dimensional hypervolume of time, frequency, amplitude and spatial direction [6]. This section documents how the base formula above is applied there; a standalone, dependency-free re-implementation of the same mechanism is provided in `../hutchinson_synthesis_application/` for reference.

4.2.1 Role in the Synthesiser

In Hutchinson Synthesis, azimuth and elevation on the sphere S^2 are treated as two effective dimensions of the sound parameter space, sampled like any other (frequency, time, amplitude) via the system’s stochastic point-process framework (GIGP). REN is layered on top of that sampling as an *acceptance-rejection filter*: it does not replace the point process governing where events tend to occur, it restricts the admissible region on the sphere to a small number of non-convex “lobes”, producing sharp source localisation with near-total exclusion between them.

4.2.2 From One Curve to One Flower per Niche

The 1D formula above defines a single boundary curve over the full angular range $[0, 2\pi)$. In Hutchinson Synthesis it is applied locally, once per acoustic niche:

1. A center (az_c, el_c) is placed on the sphere for each niche. Centers are spread apart via a repulsive relaxation on S^2 : each point is pushed away from every other point, the resulting force is projected onto the local tangent plane (to keep the point on the sphere), and the point is renormalised after each step. This produces approximately equidistant centers regardless of how many niches are active.
2. Around each center, a local tangent-plane (gnomonic) projection is built: any point (az, el) on the sphere is mapped to local polar coordinates (r, φ) relative to that center.
3. The REN boundary is evaluated in this local frame as

$$r_{\text{REN}}(\varphi) = R \cdot \exp\left(-\lambda \left|\sin\left(\frac{k\varphi}{2}\right)\right|^q\right) \quad (8)$$

where R is the niche’s radius (auto-scaled from the number of active niches, or user-defined). The angle is halved with respect to the base formula of eq. (1) so that the intended number of lobes k is recovered after the tangent-plane projection.

4.2.3 Magnet and Guaranteed Clamp

Each candidate event position is first assigned to its nearest niche center (spherical distance), then pulled toward that niche’s REN lobes with a **force** parameter $\in [0, 1]$:

$$r_{\text{final}} = r_{\text{curr}}(1 - \text{force}) + \min(r_{\text{curr}}, r_{\text{REN}}(\varphi)) \cdot \text{force} \quad (9)$$

At **force** = 0 the position is left untouched; at **force** = 1 a final clamp step guarantees the position falls on or inside the lobes, projecting it onto the nearest lobe border if it would otherwise fall outside. This is the mechanism behind the REN spatial-distribution audio and visual examples available in the **examples/** folder of the repository [6]: with high λ , sources concentrate tightly in the k lobes, producing near-total spatial silence in the excluded zones between them.

4.2.4 A Note on the Codebase

The Hutchinson Synthesis codebase also contains a separate, exploratory function (**ren_2d**) that combines independent azimuth and elevation lobe counts multiplicatively. It is not invoked by the generation pipeline and is not the mechanism described above or used to produce the published results; it is mentioned here only to avoid confusion for anyone reading that source directly.

5 Conclusion

The REN formula is a minimal yet highly flexible system for generating complex radial shapes, with direct control over both global and local structure. Its application inside Hutchinson Synthesis, described above, shows how the same compact 1D curve can be lifted onto the sphere S^2 to constrain the spatial dimension of a sound synthesiser, without changing its mathematical core.

References

- [1] G. Lamé, *Sur les surfaces isothermes*, 1838.
- [2] J. Fourier, *Théorie analytique de la chaleur*, 1822.
- [3] L. Euler, *Introductio in analysin infinitorum*, 1748.
- [4] W. Feller, *An Introduction to Probability Theory*, Wiley, 1968.
- [5] J. Gielis, “A generic geometric transformation that unifies a wide range of natural and abstract shapes”, *American Journal of Botany*, 2003.
- [6] A. Di Furia, N. Andreuccetti, *Hutchinson Synthesis* [software repository], <https://github.com/anthonydifuria/hutchinson-synthesis>, 2026.